

Data Management for Data Science



Homework 1 & 2 — SQL, Evaluation and Optimization

Italian electricity market - Terna open data - January, February & March 2026

Italian electricity market — March 2026

Open data published by Terna, the Italian transmission system operator. Nine CSV exports covering demand, generation, cross-border exchange, zonal transit limits, adequacy and FER connection requests.

FABBISOGNO

Demand per bidding zone (15-min)

total_load, forecast_total_load [19202 rows]

GENERAZIONE

Actual generation by primary source

Thermal, Wind, Hydro, PV, Geo [14402 rows]

TRASMISSIONE

Cross-border exchange with neighbours

import, export, scheduled_balance [19202 rows]

MGP

Day-ahead market transit limits

zone_from -> zone_to [29570 rows]

MI

Intra-day market transit limits

sessions CRIDA1/2/3 [67058 rows]

MSD + MSD INPUT

Balancing market margins / forecast load

sessions MSD1..MSD6 [375234 rows] [61042 rows]

ADEGUATEZZA CONSUNTIVO

Available capacity by macroarea / fuel

Nord, Sud_Isole [6050 rows]

ADEGUATEZZA PREVISIONE

Forecast capacity, per macrouser

ENEL, EDF, A2A, ... [22292 rows]

CONNESSIONI

FER grid-connection requests by region

status, capacity, num_requests [67 rows]

From raw CSV to a constrained relational schema

01

raw schema

- One table per CSV
- All columns as TEXT
- Load CSV without parse errors

02

clean schema

- Proper types: TIMESTAMP, NUMERIC, VARCHAR
- NOT NULL and CHECK on non-negative / allowed values
- Composite primary keys (date, zone, session)
- INSERT filters out the 'Applied filters' footer row

03

dimension tables + FK

- clean.zone (is_italian, is_aggregate)
- clean.source (renewable flag)
- FOREIGN KEYS from every fact table
- Enumerated sessions CRIDA1..3 / MSD1..6

Rationale: the raw layer absorbs dirty CSV input; the clean layer turns types, domains and identifiers into constraints the DBMS can actually use for joins, keys and query plans.

Conclusion

Indexes, materialization and query rewriting are the three levers that turn the slow plans into the fast ones.

Query Type	Pre-Opt Runtime	Optimization Method	Post-Opt Runtime	Cost Analysis Before	Cost Analysis After
Q3 — Net-importer countries (correlated AVG per country)	9.7 s	Materialized view + index on country	15 ms	The inner query is executed once for each tuple of the outer query	D log 2B
Q8 — Renewables at daily peak (correlated daily AVG on Italy)	2.9 s	Precomputed daily-avg table + stored date_only + indexes	19 ms	The inner query is executed once for each tuple of the outer query	D log 2B
Q9 — MSD6 vs MI anti-join (NOT EXISTS on intraday)	1.3 s	Composite B+-tree indexes	958 ms	B(D+RC)	D log 2B
Q10 — Critical forecast hours (double nested AVG)	172 ms	Materialized critical_hours + index on date	13 ms	The inner query is executed once for each tuple of the outer query	D log 2B